

Comments on the Lu-Su Authenticated Key Agreement Scheme in Ad hoc Networks

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Abstract

Lu and Su [Comm. Of the CCISA 12(1) (2006) pp.93-104] proposed an authenticated key agreement scheme in Ad hoc Networks. Their scheme based on Shamir's ID-based cryptosystem and Okamoto's key distribution system attempts concurrently to achieve two goals: (1) any mobile node A can authenticate the other one B, who is communicating with A, is illegitimate node or not, and (2) A and B can obtain a secure common session key by using key agreement protocol. Unfortunately, in this paper, we will show that the Lu-Su scheme is vulnerable to both the impersonation and the forgery attack. In addition, some modifications are provided to enhance the security problems.

1 Introduction

In recent years, mobile ad-hoc networks have attracted significant attentions because they provide anytime-anywhere networking services. A peer-to-peer wireless network formed by a collection of nodes dynamically without relying on the support from any fixed infrastructure or centralized administration is called an ad-hoc network. Nodes of a mobile ad-hoc network are often mobile such that they can create instant communication networks, which are suitable for the applications of emergency, commercial, rescue and military missions.

Security is one of the most important concerns for mobile ad-hoc networks. Generally, four fundamental security requirements are addressed: confidentiality, authentication, integrity and availability. However, by its inherent nature, such as open medium, absence of fixed central structure, dynamically changing topology and constrained capability [1-8], mobile ad-hoc networks are highly vulnerable to various security threats.

In order to achieve the security requirements above-mentioned, trust establishment, key management and authorization must be taken into the considerations. Recently, several protocols

have been proposed to solve the problem of key management over wireless and ad-hoc networks. In [9-11], the protocols proposed based on key pre-distribution. However, this assumption is only suitable for the mobile nodes with the capability of knowing other group members beforehand. In [12-14], the protocols proposed by using the concept of mobile certificate authorities (CA). In such protocols, CA is responsible for issuing a valid certificate to all the nodes. Based on the certificate, these protocols are easy to achieve the properties of data integrity, authentication and non-repudiation. The protocols in [15-16], apply an airplane or satellite to act as the role of CA, such that obtain the object of mobility to verify the public key. However, the structure is established more difficult in reality. A better solution is called ID-based key management scheme, which uses user's identity to be the user's public key. For establishing a group key, key establishment using contributory key agreement protocols are discussed in [17-19]. However, in some circumstances, there exists the requirement of any two mobile nodes securely communication with each other.

Recently, Lu and Su proposed an ID-based authenticated key agreement scheme in ad-hoc networks in [20], which is suitable for the application of any two mobile nodes securely communicating with each other. Their scheme based on Shamir's ID-based cryptosystem and Okamoto's key distribution system attempts concurrently to achieve two goals: (1) any mobile node A can authenticate the other one B, who is communicating with A, is illegitimate node or not, and (2) A and B can obtain a secure common session key by using key agreement protocol. Unfortunately, in this paper, we will show that the Lu-Su scheme is vulnerable to both the impersonation and the forgery attack. In addition, some modifications are provided to enhance the security problems.

In next section, a brief review of the Lu-Su scheme is provided. Section 3 discusses the security of Lu-Su scheme and gives an improved version. Section 4 analyzes the security of our

improved scheme. Finally, a conclusion is given.

2 Briefly review the Lu-Su Scheme

There are three entities: mobile node A, mobile node B and the trusted center TC in the Lu-Su scheme. Node A and B, who try to communicate with each other, may be any pair of mobile nodes in the Ad hoc network. TC is responsible for the authentication of the legitimacy of a mobile node. A new mobile node, wants to attend this ad hoc network, must register to the TC first. The Lu-Su scheme is based on RSA. Their scheme is composed of four phases: the initialization phase, the registration phase, the authentication phase and the key agreement phase. A detailed description presents as follows:

Initialization Phase:

TC generates the system parameters as following steps:

1. Generate two large and strong distinct primes p and q , then he computes $N = pq$.
2. Let α be the primitive element with the maximal order in the multiplicative group Z_N^* .
3. Determine a pair of public and secret keys (e, d) , satisfying $e * d \equiv 1 \pmod{\phi(N)}$ with $\gcd(e, \phi(N)) = 1$.
4. Let f is a one-way hash function. Then publish N , α , f and e public, and keep p , q , d secret.

Registration Phase:

A mobile node A submits his identification information (including name, e-mail address etc.) to TC for registering. After the TC identifies the node, it will issue an identity ID_A and a corresponding secret signing key $g_A = ID_A^d \pmod{N}$ to the node over a secure channel. Upon receiving the message from TC, node A can verify his secret signing key by the equation $ID_A = g_A^e \pmod{N}$.

Authentication Phase:

If mobile node A wants to communicate with node B, he performs the following steps (The operation of B is the same as A):

1. Choose a random number r_A .
2. Compute $y_A = g_A(\alpha)^{r_A} \pmod{N}$, $t_A = r_A^e \pmod{N}$, and $s_A = g_A(r_A)^{f(y_A, t_A, T_A, ID_B)} \pmod{N}$, where T_A is the current time.
3. A sends the message

$C_A = (ID_A, y_A, t_A, s_A, T_A)$ to node B. Similarly, B sends the message $C_B = (ID_B, y_B, t_B, s_B, T_B)$ to node A.

4. Upon receiving the message C_B from node B, the mobile node A verifies whether the following equation holds:

$$s_B^e \equiv ID_B \cdot (t_B)^{f(y_B, t_B, T_B, ID_A)} \pmod{N}.$$

If it holds, node A is convinced that the message C_B is valid and truly transmitted from the node B. Otherwise; node A will reject the communication with node B.

Key agreement Phase:

Node A computes the session key as follows:

$$\begin{aligned} SK_A &= (y_B^e / ID_B)^{r_A} \pmod{N} \\ &= \alpha^{e r_A r_B} \pmod{N} \end{aligned}$$

Similarly node B arrive at the same session key, since the following holds

$$\begin{aligned} SK_B &= (y_A^e / ID_A)^{r_B} \pmod{N} \\ &= \alpha^{e r_A r_B} \pmod{N} \end{aligned}$$

3 Cryptanalysis and improvement of the Lu-Su Scheme

In this section, we will show that the Lu-Su scheme is insecure against two attack mechanisms. In their scheme, any mobile node with identity ID_i attempts to communicate with other node, he must be able to create a valid signature message C_i . Such that, he can passes the authentication phase of the opponent. In other words, it means that any mobile node must register to the TC first in order to obtain a pair of (ID_i, g_i) and then he can make a valid signature with his signing key g_i . However, the following two attacks can make the evil mobile node create other valid pair (ID_v, g_v) without registering to the TC. Therefore, the evil mobile node can impersonate another legal node to communicate with other nodes.

3.1 First impersonation attack

Any legal mobile node can easily to impersonate other legal mobile node in the scheme. For instance, suppose that an evil mobile node with a pair of (ID_i, g_i) plans to create another valid pair of (ID_v, g_v) such that the pair fits to $g_v = ID_v^d \pmod{N}$.

The evil mobile node cannot compute the signing key g_v directly since he doesn't know the secret key d of the TC. However, he can find both the valid identity ID_v with specific format

and the signing key g_v from the following equations:

$$ID_v = ID_i^k \bmod N$$

where k is an arbitrary integer to make ID_v satisfy the specific format and the corresponding signing key g_v can be obtained by:

$$\begin{aligned} g_v &= ID_v^d \bmod N \\ &= (ID_i^k \bmod N)^d \bmod N \\ &= g_i^k \bmod N \end{aligned}$$

Once the pair of (ID_v, g_v) is obtained, the evil mobile node with identity ID_i can impersonate the mobile node with identity ID_v to sign the signature $C_v = (ID_v, y_v, t_v, s_v, T_v)$ and pass the authentication phase in the scheme.

3.2 Second impersonation attack

If some intruders (legal mobile nodes with identities $ID_{ij}, j=1,2,\dots$) cooperate to obtain a valid pair of (ID_v, g_v) as follows:

$$ID_v = \prod ID_{ij} \bmod N$$

Suppose that ID_v fulfills the identity format, its corresponding signing key can be computed by:

$$\begin{aligned} g_v &= ID_v^d \bmod N \\ &= (\prod ID_{ij} \bmod N)^d \bmod N \\ &= (\prod g_{ij} \bmod N) \bmod N \end{aligned}$$

3.3 The forgery attack

The intruder can easily forge a valid signature $C_v = (ID_v, y_v, t_v, s_v, T_v)$ to cheat the opponent. Assumed that the intruder wants to communicate with the mobile node with identity ID_B , he first sets (y_v, t_v, s_v) to be some random integers and chooses the desired time T_v . After that, he computes the corresponding identity ID_v from the equation:

$$ID_v = s_v^e / (t_v)^{f(y_v, t_v, T_v, ID_B)} \bmod N.$$

Now, the signature $C_v = (ID_v, y_v, t_v, s_v, T_v)$ can easily pass the authentication phase of the mobile node with identity ID_B . Note that, anyone may act as the intruder. He is not necessary to be a legal mobile node. From this point view, it seems to be a more serious security leak in the Lu-Su scheme.

3.4 Improvement

Some modifications are provided to improve the security of the Lu-Su scheme. We only modify the registration phase and the key agreement phase in the scheme.

In registration phase, the operation of TC for computing a secret signing key g_A of mobile node with identity ID_A is replaced with the

equation as follows.

$$g_A = (f(ID_A))^d \bmod N$$

In key agreement phase, two steps are modified as:

1. Verify whether the following equation holds:

$$s_B^e \equiv f(ID_B) \cdot (t_B)^{f(y_B, t_B, T_B, ID_B)} \bmod N$$

If it is not, the communication request is rejected.

2. Compute the session key

$$SK_A = (y_B^e / f(ID_B))^{r_A} \bmod N.$$

4. Security analysis of the improved scheme

We demonstrate that the improved scheme is secure and amended the security flaws mentioned in section 3.

- Attack scenario 1: The intruder, a legal mobile node, with a pair of (ID_i, g_i) wants to directly deduce another valid pair of (ID_v, g_v) without registering to the TC.

The intruder may try to use a random value k to compute the signing key $g_v = g_i^k \bmod N$. However, he cannot compute the corresponding identity ID_v . Because of solving the equation of $f(ID_v) = f(ID_i)^k \bmod N$, the intruder must face the difficulty of reversing the one-way hash function.

- Attack scenario 2: The intruders, some legal mobile nodes, want to cooperatively deduce another valid pair of (ID_v, g_v) without registering to the TC.

The intruders may compute the signing key $g_v = (\prod g_{ij}) \bmod N$ by multiplying their signing key g_{ij} together. However, they cannot compute the corresponding identity ID_v . If the intruders want to compute the corresponding identity ID_v from the equation $f(ID_v) = \prod f(ID_{ij}) \bmod N$, they still face the difficulty of reversing the one-way hash function.

- Attack scenario 3: The intruder tries to perform our forgery attack in section 3.3 on the improved scheme.

The intruder first sets (y_v, t_v, s_v) to be some random values and chooses a desired time T_v and then he tries to solve the corresponding identity ID_v from the equation $f(ID_v) = s_v^e / (t_v)^{f(y_v, t_v, T_v, ID_B)} \bmod N$. However, he must face the difficulty of reversing the one-way hash function.

- Attack scenario 4: After fabricating an identity ID_v with valid format, the intruder tries to forge a valid message $C_v = (ID_v, y_v, t_v, s_v, T_v)$ to pass the authentication phase.

Without knowing the signing secret key $g_v = ID_v^d \bmod N$, the intruder is difficult to compute the set of values (y_v, t_v, s_v) from the equation $s_v^e = f(ID_v)(t_v)^{f(y_v, t_v, T_v, ID_B)} \bmod N$. Since he will face the difficulty of reversing the one-way hash function or factorization problems [21].

4. Conclusion

In this paper, we described the security problems of the Lu-Su scheme in Ad hoc networks. Their scheme is vulnerable to both the impersonation attack and the forgery attack. In order to amend above flaws, we make an improvement to the Lu-Su scheme. In addition, detailed security analyses of the improved scheme are discussed. Therefore, the improved scheme can achieve the requirements of identity authentication and session key generation in Ad hoc networks.

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